

SNOWPRO® ADVANCED: DATA SCIENTIST (DSA-C03) EXAM STUDY GUIDE

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SNOWPRO ADVANCED: DATA SCIENTIST STUDY GUIDE OVERVIEW

This is a self-learning study guide that highlights concepts that may be covered on Snowflake's SnowPro Advanced: Data Scientist Certification exam.

This study guide does not guarantee certification success.

Holding the SnowPro Core certification in good standing is a prerequisite for taking the Advanced: Data Scientist certification.

For an overview and more information on the SnowPro Core Certification exam or SnowPro Advanced Certification series, please navigate [here](#).

RECOMMENDATIONS FOR USING THE GUIDE

This guide will show the Snowflake topics and subtopics covered on the exam. Following the topics will be additional resources consisting of videos, documentation, blogs, or exercises to help you understand the Data Scientist role on the Snowflake AI Data Cloud.

Estimated length of study guide: 10 – 13 hours

Some links may have more value than others, depending on your experience, the same amount of time should not be spent on each link. Some links may appear in more than one domain.

SNOWPRO ADVANCED: DATA SCIENTIST CERTIFICATION OVERVIEW

The **SnowPro Advanced: Data Scientist** exam tests advanced knowledge and skills used to apply comprehensive data science principles, tools, and methodologies using Snowflake. The exam will assess skills through scenario-based questions and real-world examples.

This certification will test the ability to:

- Outline data science concepts
- Implement Snowflake data science best practices
- Prepare data and feature engineering in Snowflake
- Train and use machine learning models
- Use data visualization to present a business case (e.g., model explainability)
- Implement model lifecycle management
- Use GenAI and LLM capabilities in Snowflake

Target Audience:

2+ years of practical data science experience with Snowflake, in an enterprise environment. In addition, successful candidates may have:

- A statistical, mathematical, or data science education (or equivalent work experience)
- Background working with one or more of the following programming languages (e.g., Python, SQL, Scala, etc.)
- Experience modeling and using machine learning platforms (e.g., SageMaker, Azure Machine Learning, GCP AI platform, AutoML tools, etc.)
- An understanding of various open source and commercial frameworks and libraries (e.g., scikit-learn, TensorFlow, etc.)
- Experience preparing, cleaning, and transforming datasets from multiple sources
- Experience creating features for machine learning training
- Experience training, validating, and interpreting models
- Experience putting a model into production and monitoring the model in production
- Experience presenting data using visualization tools
- Leveraging GenAI and LLM methodologies (e.g., RAG, prompting, fine tuning, task-specific models such as Document AI and Snowflake Copilot)

This exam is designed for:

- Data Scientists
- AI or ML Engineers

SNOWPRO ADVANCED: DATA SCIENTIST PREREQUISITE

Eligible individuals must hold an active SnowPro Core Certified credential. If you feel you need more guidance on the fundamentals, please see the [SnowPro Core Exam Study Guide](#).

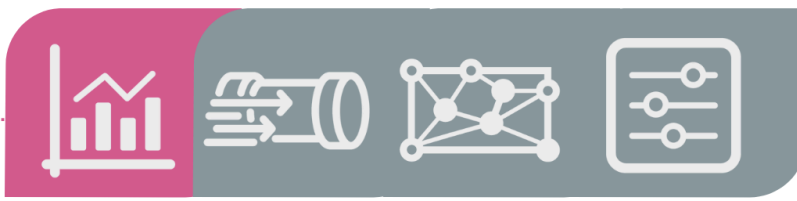
STEPS TO SUCCESS

1. [Review the Data Scientist Exam Guide](#)
2. [Attend Snowflake's Instructor-Led Data Scientist Training](#)
3. [Review and study applicable white papers and documentation](#)
4. Get hands-on practical experience with relevant business requirements using Snowflake
5. [Attend Snowflake Webinars](#)
6. [Attend Snowflake Virtual Hands-on Labs to gain practical experience](#)
7. Practice with sample questions [here](#)
8. [Schedule your exam](#)
9. Take your exam!

SNOWPRO ADVANCED: DATA SCIENTIST SUBJECT AREA BREAKDOWN

This exam guide includes test domains, weightings, and objectives. It is not a comprehensive listing of all the content that will be presented on this examination. The table below lists the main content domains and their weightings.

Domain	Domain Weightings on Exams
1.0 Data Science Concepts	17%
2.0 Data Preparation and Feature Engineering	27%
3.0 Model Development	31%
4.0 Model Deployment	25%



Domain 1.0: Data Science Concepts

1.1 Define machine learning concepts for data science workloads.

- Machine Learning
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning

1.2 Identify machine learning problem types.

- Supervised Learning
 - Structured Data
 - Linear regression
 - Binary classification
 - Multi-class classification
 - Time-series forecasting
 - Unstructured Data
 - Image classification
 - Segmentation
- Unsupervised Learning
 - Clustering
- GenAI
 - Association models

1.3 Summarize the machine learning lifecycle.

- Data collection
- Data visualization and exploration
- Feature engineering
- Training models
- Model deployment
- Model monitoring and evaluation (e.g., model explainability, precision, recall, accuracy, confusion matrix)
- Model versioning

1.4 Define statistical concepts for data science.

- Normal versus skewed distributions (e.g., mean, outliers)
- Central limit theorem
- Z and T tests
- Bootstrapping
- Confidence intervals

Domain 1.0 Study Resources

Lab Guides

[Attaining Consumer Insights with Snowflake and Microsoft Power BI](#)

Additional Assets

[Complete Guide to Machine Learning](#)

[Evaluation Metrics](#) (blog)

[Confidence Intervals](#) (paper)

[How to Accelerate Data Science with Snowflake](#) (video)

[Optimizing the Machine Learning Lifecycle and MLOps](#) (white paper)

[Performance Metrics in Machine Learning \[Complete Guide\]](#) (blog)

[Propensity Modeling: Using Data \(and Expertise\) to Predict Behavior](#) (blog)

[Reinforcement Learning 101](#) (blog)

[Specific ML Problem Types: GenAI](#)

[The Machine Learning Lifecycle](#) (blog)

[Understanding Confusion Matrix](#) (blog)

[Customer Churn Prediction Using Machine Learning](#) (blog)

[Regression: Definition, Analysis, Calculation, and Example](#) (blog)

[The Complete Guide to Time Series](#)

[Forecasting Models](#) (blog)

[What Is Variance in Statistics? Definition, Formula, and Example](#) (blog)



Domain 2.0: Data Preparation and Feature Engineering

2.1 Prepare and clean data in Snowflake.

- Use Snowpark for Python and SQL
 - Aggregate
 - Joins
 - Identify critical data
 - Remove duplicates
 - Remove irrelevant fields
 - Handle missing values
 - Data type casting
 - Sampling data

2.2 Perform exploratory data analysis in Snowflake.

- Snowpark and SQL
 - Identify initial patterns (i.e., data profiling)
 - Connect external machine learning platforms and/or notebooks (e.g., Jupyter)
- Use Snowflake native statistical functions to analyze and calculate descriptive data statistics.
 - Window Functions
 - MIN/MAX/AVG/STDEV
 - VARIANCE
 - TOPn
 - Approximation/High Performing function
- Linear Regression
 - Find the slope and intercept
 - Verify the dependencies on dependent and independent variables

2.3 Perform feature engineering on Snowflake data.

- Preprocessing
 - Scaling data
 - Encoding
 - Normalization
- Data Transformations
 - DataFrames (i.e., pandas, Snowpark, Snowpark pandas)
 - Derived features (e.g., average spend)
- Binarizing data
 - Binning continuous data into intervals
 - Label encoding
 - One hot encoding
- Snowpark Feature Store

2.4 Visualize and interpret the data to present a business case.

- Statistical summaries
 - Snowsight with SQL
 - Interpret open-source graph libraries
 - Identify data outliers
- Snowflake Notebooks

Domain 2.0 Study Resources

Lab Guides

[Accelerating Data Teams with dbt Cloud & Snowflake](#)
[Feature Engineering \(workload guide\)](#)
[Snowpark Python Demo \(lab\)](#)
[Advanced Guide to Snowflake Feature Store \(lab\)](#)

Additional Assets

[6 Critical Reasons to Normalize Data \(blog\)](#)
[Data Discovery Made Easy with Snowsight \(blog\)](#)
[How Post Holdings Embraces Growth and Innovation with Snowflake for Data Science \(video, requires email\)](#)
[How to Build a Foundation for Data Science with Snowflake's Data Cloud \(video\)](#)
[How, When, and Why Should You Normalize / Standardize / Rescale Your Data? \(blog\)](#)
[How to Choose a Feature Selection Method For Machine Learning \(blog\)](#)
[Linear Regression in Snowflake \(blog\)](#)
[Next-Generation Automated Feature Engineering with DataRobot and Snowflake \(paper\)](#)
[SMOTE for Imbalanced Classification with Python \(blog\)](#)
[Undersampling Algorithms for Imbalanced Classification \(blog\)](#)
[Missing Data Imputation \(blog\)](#)
[Does SVM Need Feature Scaling Or Normalization? \(blog\)](#)

Snowflake Documentation Links

[Drivers](#)
[Load Data into Snowflake](#)
[Snowpark API](#)
[Snowsight](#)
[SQL Command Reference](#)

SQL Function Reference

[pandas on Snowflake](#)
[APPROX_COUNT_DISTINCT](#)
[Data Transformation and Dimensionality Reduction](#)
[Primary Data Transformation and Encoding](#)
[Feature Store Overview](#)
[Feature Store API](#)
[snowflake.ml.modeling](#)
[Functions](#)
[Aggregate functions](#)
[DataFrame](#)
[Working with DataFrames in Snowpark Python](#)
[Import Python packages to use in notebooks](#)
[DECODE](#)
[NTILE](#)
[Visualizing worksheet data](#)
[Querying data using worksheets](#)
[About Snowflake Notebooks](#)
[Schedule your Snowflake Notebook to run](#)



Domain 3.0: Model Development

3.1 Connect data science tools directly to data in Snowflake.

- Connecting Python to Snowflake
 - Snowpark
 - Snowpark ML
 - Python connector with Pandas support
- Connecting from external IDE (e.g., Visual Studio Code)
- Snowpark languages

3.2 Leverage GenAI and LLM models in Snowflake.

- Snowflake Cortex
 - Vector embedding
 - Prompt engineering
 - Fine tuning
 - Task-specific models (e.g., categorization, summarization, sentiment analysis, information extraction)

3.3 Train a data science model.

- Build a data science pipeline
 - Automation of data transformation (e.g., dynamic tables)
 - Python User-Defined Functions (UDFs)
 - Python User-Defined Table Functions (UDTFs)
 - Python stored procedures

- Hyperparameter tuning
- Optimization metric selection (e.g., log loss, AUC, RMSE)
- Partitioning
 - Cross validation
 - Train validation hold-out
- Down/up-sampling
- Training with Python stored procedures
- Training outside Snowflake through external functions
- Training with Python User-Defined Table Functions (UDTFs)

3.4 Validate a data science model.

- ROC curve/confusion matrix
 - Calculate the expected payout of the model
- Regression problems
- Residuals plot
 - Interpret graphics with context
- Model metrics

3.5 Interpret a model.

- Feature impact
- Partial dependence plots
- Confidence intervals
- SHAP values

Domain 3.0 Study Resources

Lab Guides

[Data Science Pipeline](#)

Additional Assets

[12 Important Model Evaluation Metrics for Machine Learning Everyone Should Know](#) (blog)

[A Gentle Introduction to k-fold Cross-Validation](#) (blog)

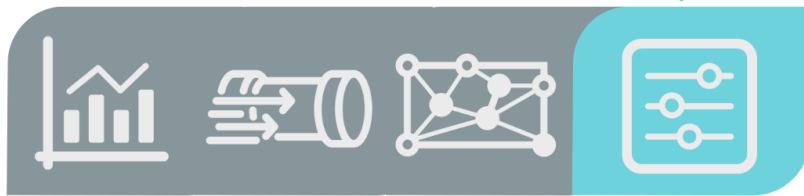
[A Look at Precision, Recall, and F1-Score](#) (blog)

[Classification: ROC Curve and AUC](#) (instruction)

[Common Metrics for Time Series Analysis](#) (blog)

[Metrics To Evaluate Machine Learning Algorithms in Python \(blog\)](#)
[Partial Dependence and Individual Conditional Expectation Plots \(instruction\)](#)
[Receiver Operating Characteristic \(ROC\) Curves \(instruction\)](#)
[Residual Plot: Definition and Examples \(blog, video\)](#)
[What are Residuals? \(blog\)](#)
[What is ML model governance? \(blog\)](#)
[Snowflake Service Consumption Table \(Paper\)](#)
[Distributed ML with Snowpark Python UDFs \(blog\)](#)
[Common Loss functions in machine learning \(blog\)](#)
[What Is a Confidence Interval and How Do You Calculate It? \(blog\)](#)
[Create Your Own Coefficient Plot Function in Python \(blog\)](#)
[Welcome to the SHAP documentation \(instruction\)](#)
Snowflake Documentation Links
[Databases, Tables and Views](#)
[Data Governance in Snowflake](#)
[Drivers](#)
[External Functions](#)
[Masking Policies](#)
[Snowpark API](#)

[SQL Command Reference](#)
[Introduction to external tables](#)
[Using pandas DataFrames with the Python Connector](#)
[Python Connector API](#)
[Large Language Model \(LLM\) Functions \(Snowflake Cortex\)](#)
[SUMMARIZE \(SNOWFLAKE.CORTEX\)](#)
[Fine-tuning \(Snowflake Cortex\)](#)
[EMBED_TEXT_768 \(SNOWFLAKE.CORTEX\)](#)
[snowflake.cortex.Complete](#)
[CREATE TASK](#)
[Stored procedures overview](#)
[SYSTEM\\$STREAM_HAS_DATA](#)
[Choosing whether to write a stored procedure or a user-defined function](#)
[User-defined functions overview](#)
[Training Machine Learning Models with Snowpark Python](#)
[Writing Snowpark Code in Python](#)
[Worksheets](#)
[Designing Python UDFs](#)
[Model Explainability](#)



Domain 4.0: Model Deployment

4.1 Move a data science model into production.

- Use an external hosted model
 - External functions
 - Pre-built models
- Deploy a model in Snowflake
 - Vectorized/Scalar Python User-Defined Functions (UDFs)
 - Pre-built models
 - Storing predictions
 - Stage commands
 - Snowflake Model Registry
 - Model logging and retrieving
 - Snowpark Container Services

4.2 Determine the effectiveness of a model and retrain if necessary.

- Metrics for model evaluation
 - Data drift /Model decay
 - Data distribution comparisons (Do the data making predictions look similar to the training data? Do the same data points give the same predictions once a model is deployed?)
- Area under the curve
- Accuracy, precision, recall
- RMSE (regression)

4.3 Outline model lifecycle and validation tools.

- Metadata tagging
- Model versioning with Snowflake Model Registry
- Automation of model retraining

Domain 4.0 Study Resources

Lab Guides

[Accelerate ML Feature Engineering with Snowflake and Amazon SageMaker Data Wrangler](#)

[Snowpark Python Top Three Tips for Optimal Performance](#)

[Snowflake Model Registry: Operationalizing Model Management Guide](#)

Additional Assets

[Accelerating Model Velocity through Snowflake Java UDF integration \(blog\)](#)

[Enabling Scalable Machine Learning in Snowflake using SQL, Python and Bodo \(blog\)](#)

[External Functions Tutorial \(video\)](#)

[How to Detect Model Drift in MLOps Monitoring \(blog\)](#)

[How to Detect Model Drift in ML Monitoring \(blog\)](#)

[In Place Machine Learning in Snowflake \(blog\)](#)

[Taking AI on Snowflake to the Next Level with Dataiku, Java UDFs & Snowpark \(blog\)](#)

[Transform Data in Snowflake Data Cloud with Java UDFs in Tellius \(blog\)](#)

[Understanding AUC - ROC Curve \(blog\)](#)

[Regression Metrics for Machine Learning \(blog\)](#)

[Understanding and detecting different types of drift \(blog\)](#)
[What Is a Confusion Matrix And How to Read It? \(blog\)](#)

Snowflake Documentation Links

[External Functions](#)
[Functions and Procedures](#)

[SQL Command Reference](#)
[Snowpark API](#)
[Designing Java UDFs](#)
[Vectorized Python UDFs](#)
[Snowflake Model Registry](#)
[Model commands](#)
[Classification \(Snowflake ML Functions\)](#)

SNOWPRO ADVANCED: DATA SCIENTIST SAMPLE QUESTIONS

1. A Data Scientist is building a machine learning model, leveraging Python and SQL. The Data Scientist will need to present the initial results using Streamlit visualizations and must be able to quickly implement changes and run individual queries while presenting.

Which Snowflake service or feature will meet these requirements MOST efficiently?

- A. Snowsight Dashboards
 - B. Snowflake Notebooks*
 - C. Snowflake Python Worksheets
 - D. Snowflake SQL Worksheets
2. What is the correct sequence of activities that need to be performed when running a data science workload?
- A. Data Collection > Feature Engineering and Transformation > Data Visualization, Exploration and Understanding > Model Deployment > Model Training > Model Monitoring
 - B. Data Collection > Data Visualization, Exploration and Understanding > Feature Engineering and Transformation > Model Training > Model Monitoring > Model Deployment
 - C. Data Visualization, Exploration and Understanding > Data Collection > Feature Engineering and Transformation > Model Deployment > Model Monitoring > Model Training
 - D. Data Collection > Data Visualization, Exploration and Understanding > Feature Engineering and Transformation > Model Training > Model Deployment > Model Monitoring*
3. Which estimating function is used to identify the approximate number of times a value appears in a data set?
- A. APPROX_COUNT_DISTINCT
 - B. APPROX_PERCENTILE
 - C. APPROX_TOP_K*
 - D. APPROX_SIMILARITY

4. A Data Scientist at Snowbear Airlines has trained a classification model for predicting the overbooking of a flight. The Data Scientist sees the following for one of the flights:

The flight was not overbooked, and the model classified the flight as not overbooked.

On a confusion matrix, what is the correct category of this description?

- A. True Negative*
 - B. True Positive
 - C. False Negative
 - D. False Positive
5. What is the correct way to interpret a linear regression model that has an R^2 of .85?
- A. The model predicts that response variable will change by approximately .85 as the explanatory variable changes by 1.
 - B. If the model was repeated multiple times it would capture the true slope 85% of the time in various intervals.
 - C. There is a strong positive correlation between the response and explanatory variables.
 - D. 85% of the variability in the response variable can be explained by the linear association between the response and explanatory variables.*

NEXT STEPS

REGISTERING FOR YOUR EXAM

When you are ready to register for the exam navigate [here](#) to get started. Select the exam you want to take and click “Register Now”. This will take you to our Certification Management system where you will register to take the exam.

MAINTAINING YOUR CERTIFICATION

All Snowflake Certifications expire two (2) years after your certification issue date.

SnowPro Certifications can now be recertified through the Snowflake Continuing Education (CE) program which includes these options -

- Completion of eligible Snowflake [Instructor Led \(ILT\) Training Courses](#)
- Earning of an equivalent or higher-level SnowPro Certification

Note: You must have a valid Certification to participate in the Continuing Education (CE) program.

The information provided in this study guide is provided for your purposes only and may not be provided to third parties.

IN ADDITION, THIS STUDY GUIDE IS PROVIDED “AS IS”. NEITHER SNOWFLAKE NOR ITS SUPPLIERS MAKES ANY OTHER WARRANTIES, EXPRESS OR IMPLIED, STATUTORY OR OTHERWISE, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY, TITLE, FITNESS FOR A PARTICULAR PURPOSE OR NONINFRINGEMENT.